

# In the Zone! — Controlling and Visualising Sound Zones

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## ABSTRACT

Sound zones enable multiple simultaneous sound experiences in the same physical room without interference. In this paper, we present an interactive sound zone setup that can produce two sound zones within a confined space. Through a tangible remote controller, users can change the volume, size, and position of these sound zones. In addition, we have built a custom visualisation display that provides real-time feedback of the sound zones to support users' understanding. Sound zone systems pose novel challenges for the HCI community, including how users may understand and interact with sound zones. Our setup offers a concrete solution into investigating these challenges.

## CCS CONCEPTS

• Human-centered computing → Interaction techniques.

## KEYWORDS

Sound zones, sound visualisation, ubiquitous computing, sound interaction, visualisation

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## 1 INTRODUCTION

Noise and unwanted sound pose an increasing challenge for the health and well-being in our everyday lives. This challenge is further exacerbated by the vast number of sound reproducing devices in homes, work, and public environments. Unwanted sound introduces several negative effects [12] and can affect our psychological health, with prior work pointing to psychiatric disorders and decreased psycho-social well-being as a result [27]. Wanted sound has shown to improve quality of life and health [31], including

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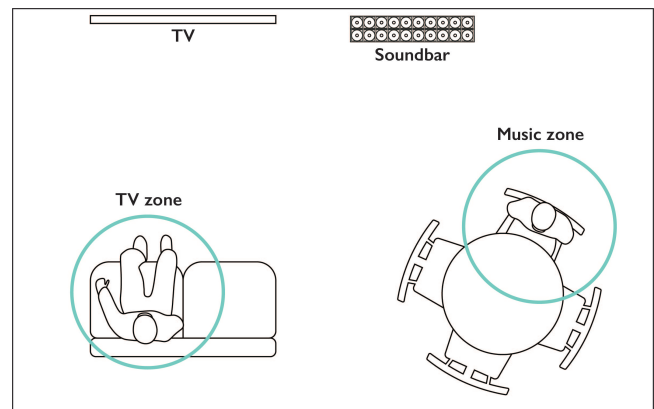


Figure 1: Illustration of two people sharing a space each with their own sound zone. From [14].

depression [23] and somatic disorders [2, 8], and improves general well-being, e.g., stress reduction and sleep betterment [3, 15]. Sound zones is an emerging technological solution to reduce unwanted sound and promote wanted sound. Loudspeaker arrays are used to create distinct areas of sound within a shared environment. This provides users with personal listening experiences without the use of headphones, and thereby still allows for social interaction. Even though sound zone research and technology have existed for years, sound zones only recently reached a level of maturity for real-world adoption. Sound zones can prove useful in contexts such as homes with multiple co-located family members listening to different content, in shared office spaces, and in a health context where alarms and other noise pose a stressful environment for patients, to give a few examples.

In a home scenario, one family member might listen to music at the dining table, while another family member watches television in the couch area (see Figure 1). This can result in an environment with overlapping sound and a bad experience for both parties [21]. In this example, sound zones may prevent the family members from hearing the sound from other sound zones. Sound zone enables users to modify their sound environment in new ways, as sound zone properties such as volume, size, and position can be controlled. A challenge for interacting with sound zones is that they are invisible and intangible for users [18], which means that users lack

knowledge of the current number of zones present as well as their characteristics such as volume, size, and position.

In this interactivity paper, we present a sound zone demonstration setup with a soundbar, tangible controllers, and a visualisation bar. In this demonstration, users receive an in-person experience of this novel technology, with two active sound zones playing a different music track. Users can interact with the sound zones using the tangible controllers by adjusting their volume, size, and position. The interaction is supported by a custom-built ultra-wide display which visualises the sound zone properties for all attendees.

## 2 RELATED WORK

### 2.1 Sound Zone Technology

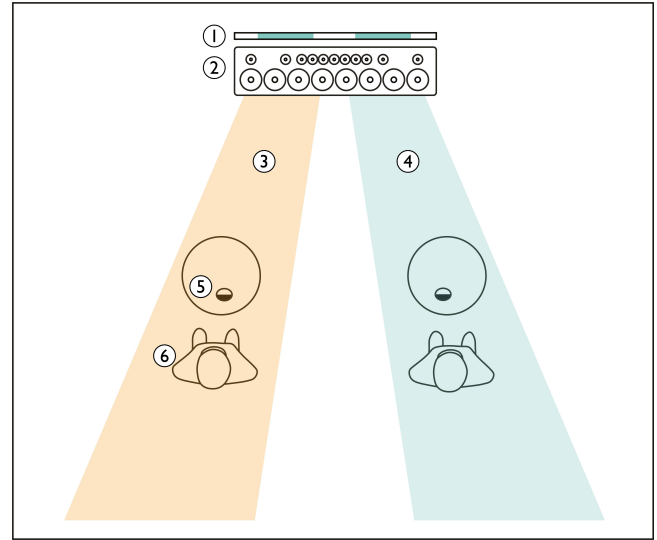
The goal of sound zones is to give multiple users a personal sound experience within the same space with minimal interference [1]. A sound zone can be defined as a physical space in which a sound is reproduced while reducing the sound pressure level of the same sound in other places [9]. The concept was initially demonstrated at the Illinois Institute of Technology in 1967 [1] but has been further developed within sound engineering and acoustics research, particularly in the last two decades. Microsoft's 'Personal Audio Space' [19] used an array of speakers to direct a sound towards a user while bystanders received a reduced volume, demonstrating a method for creating a single sound zone. For creating multiple sound zones, one method is to heighten the acoustic contrast between two or more sounds, maximising a sound within a defined area while reducing it everywhere else [4, 6]. Research into personal sound has been further extended in various application areas, including office spaces [1], car cabins [5], outdoor concerts [13], and even small mobile devices [10].

### 2.2 Sound Zone Interaction

Lundgaard et al. frame sound zones as an intervention into a person's soundscape, as a sound zone is experienced with all other sounds that are present [18]. Furthermore, they identify eight challenges for interaction design for sound zones. The eight challenges revolve around the invisibility and intangibility of sound zones, considerations of the social aspects of interacting with sound zones, and users' mental model of a sound zone. In another study, Lundgaard & Nielsen highlight how different situations result in different conceptions of what a sound zone is and which aspects of the sound zone users want to control [16]. Preliminary design workshops identified four interaction design approaches for sound zone interaction; tangible representations, light projections, familiar objects, and handheld devices [17]. This prior work offers a foundation for taking the necessary next step and develop concrete solutions for interacting with sound zones.

## 3 SOUND ZONE INTERACTIVITY SETUP

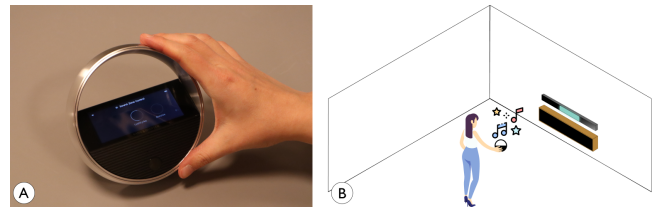
We now present the complete setup for our interactivity demonstration. The setup includes a soundbar, tangible controllers and a visualisation bar (see Figure 2). Our setup can thereby provides the attendees with the novel experience of sound zones, with interaction possibilities to change the properties of sound zones, while being visualised in real-time.



**Figure 2: The setup overview. 1) Visualisation bar, 2) Soundbar, 3) Sound zone A, 4) Sound zone B, 5) Tangible controller, and 6) User in a sound zone**

### 3.1 Soundbar

For creating the dynamic sound zones, we use a custom-built soundbar developed by Soundfocus [30]. It utilises complex sound engineering algorithms and sound processing in order to focus two different sources of sound into two separate locations within the same confined space. This allows for listening experiences with little to no overlap between multiple users' sound zones. Each sound zone is dynamic, meaning that the positions for optimal listening can be controlled in real-time to accommodate the locations of the users. The soundbar comprises of 16 driver units of 2". Each driver is fed by an individual processing channel and 15W amplifier, resulting in ample aggregated soundlevel capacity for the entire soundbar. The soundbar is meant to reproduce directionally controlled sound zones in the frequency range between 100 and 15000 hertz.



**Figure 3: A) One of the two Bang & Olufsen 'BeoRemote Halo' controllers. A capacitive touch screen is mounted inside the ring, and the ring itself is turn-able for adjusting sound zone properties. B) Illustration of the visualisation bar mounted above the soundbar.**

### 3.2 Tangible Controllers

For controlling the sound zone properties, we developed a custom application for sound zone interaction for two Bang & Olufsen

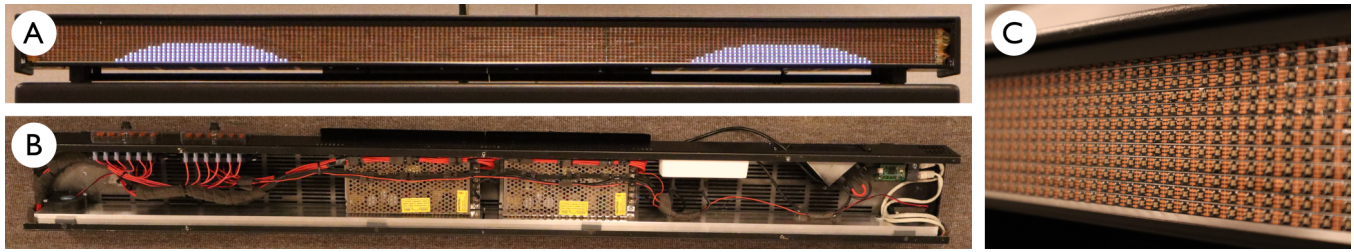


Figure 4: A) Front-side of the visualisation bar showing two sound zones. B) Internal components of the visualisation bar. C) Close-up of the LED display.

‘BeoRemote Halo’ controllers [22]. Each controller (see Figure 3-A) enables control of one sound zone, in terms of volume, size, and position. Through the touch screen on the controller, the user can select between control of these three properties. When a property is selected, the user can turn the large ring and either adjust the volume up or down, increase or decrease the size, or move the position of the sound zone left or right, respectively.

| Spatial property | Position | Volume | Size |
|------------------|----------|--------|------|
| Visualisation    |          |        |      |

Figure 5: Behaviour of the visualisation bar when visualising volume, size, and position

### 3.3 Visualisation Bar

We provide users with a visualisation of the volume, size, and position of sound zones through an ultra-wide display (inspired by e.g., [7, 11, 24, 28]) positioned directly above the soundbar (Figure 3-B). By moving and changing two half-ellipsoid shapes across the display (Figure 5), we inform the users of the spatial properties of the two active sound zones in real-time. The goal of the display is to provide information to the user with a low level of obtrusion. The shapes map the spatial properties from the three-dimensional environment onto a two-dimensional plane. This visualisation is shown on the display facing towards the listener. For example, when the sound zone is positioned at the far left of the listening area, an half-ellipsoid is shown at the far left of the display (see Figure 5). Correspondingly, the current size of the sound zone is mapped to the width of the visualisation, and the height of the half-ellipsoid represents the current volume.

**3.3.1 Implementation.** We built the ultra-wide display using very small RGB LED-strips. With 12 rows and 192 LEDs in each row, the display includes 2304 LEDs (see Figure 4). The LEDs are powered by two 30A power supplies and are controlled through a single TEENSY 4.0 development board [25] that updates the entire display at a rate of 60Hz. A Raspberry PI 4b [26] runs a NodeJS [20] server that is connected to the soundbar for synchronised feedback of the sound zones through web-sockets (socketIO library [29]).

## 4 DISCUSSION AND FUTURE WORK

In this paper, we present a setup that demonstrates possible ways to interact and understand sound zones. Sound zones show great promise in enabling people to have personal listening experiences in many contexts without down prioritising social interaction. Simultaneously, however, sound zones pose several challenges for designers and HCI researchers [18]. Most importantly, how people might interact with sound zones and understand the properties introduced by sound zones remains an open research question.

Particularly, we see two important directions for future HCI research in sound zones. First, sound zones contradict users’ understanding of how sound typically behaves. *I.e.*, for conventional sound systems, people expect that any present sound is audible as long as the volume is at a sufficient level. This changes with sound zone systems, where multiple zones can be in the same room. If the sound zones do not overlap, they are inaudible to non-users. Second, sound zones allow users to modify their sound environment by directly interacting with the sound zone properties, and thereby shape their sound experiences on a more personal level. This opens the opportunity for future research into how people may optimally interact with sound zones, and shape them to fit a plethora of use situations with multiple simultaneous users and zones. In our CHI 2022 paper, we explore how a projection-based visualisation and a display-based visualisation help users understand and interact with sound zones.

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